PETRA ANDHIKA NATANAEL

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EDUCATION

Universitas Sanata Dharma - Yogyakarta, Indonesia **Sep 2022 - Sep 2026**
Bachelor of Informatics, GPA 3.57/4.00

WORK EXPERIENCE

AI Engineer Intern | Beehive Drone **Feb 2026 - May 2026**

Computer Vision - Oil Palm Flower Ripeness Classification

- Labeled image data (Roboflow) for 4 oil palm flower classes, then built and benchmarked detection/classification models using YOLOv8, YOLOv11, and YOLO26, achieving a best accuracy of 92.6%.
- Deployed the model to a Raspberry Pi 5 with an IMX500 AI camera for drone integration, enabling real-time field detection.

Computer Vision - Barcode, QR Code & Expiry Date Detection System for Warehouse Operations

- Built a multi-threaded system (Python, OpenCV, YOLO) for real-time barcode/QR code and expiry date detection, with decoding via pyzbar/zxing-cpp and OCR (EasyOCR, PaddleOCR, Tesseract) enhanced with image preprocessing (CLAHE, super-resolution).
- Stored results in a CSV database and displayed them on a UI panel, designed for integration with a drone camera to speed up warehouse stock checks

ACADEMIC PROJECT

Shuttlecock Tracking in Badminton Videos Using TrackNetV3 | **Aug 2026**
Undergraduate Thesis (Final Project)

- Implemented TrackNetV3 (Tracking Module: U-Net Conv2D + Rectification Module: InpaintNet Conv1D) with sequence-length variation analysis (4, 6, 8, 10 frames) on 201 amateur and professional badminton rally videos.
- Conducted 4 progressive experimental scenarios (baseline, +background estimation, +Mixup, +Rectification Module); the complete pipeline with a sequence length of 8 frames delivered the most balanced performance: 91.75% accuracy, 94.11% recall, and 95.11% F1-score.

ACHIEVEMENTS

Selected for University Funding - Student Creativity Program (PKM) **May 2025 - Jul 2025**

- Proposal "Smart Parking Management System Based on Fuzzy Logic and Optical Character Recognition for Vehicles" was selected, demonstrating the ability to design an intelligent system integrating Fuzzy Logic for decision-making and OCR for vehicle license plate recognition.

ORGANIZATIONAL EXPERIENCE

Communication & Information Staff | BEM Universitas Sanata Dharma **Dec 2023 - Dec 2024**

- Managed the planning, production, and publication of visual content (photo/video) using Figma, Canva, and CapCut for the organization's digital communications.
- Managed and redesigned the organization's website using WordPress, improving usability and information timeliness.

SKILLS

- Hard Skills: Python, Java, SQL | Computer Vision & Deep Learning (OpenCV, YOLO, PyTorch, TensorFlow) | OCR (EasyOCR, PaddleOCR, Tesseract) | Edge AI Deployment (Raspberry Pi 5, IMX500)
- Soft Skills: Analytical Problem-Solving, Initiative & Proactiveness, Teamwork, Time Management, Technical Documentation